Managerial Economics

Lecture 2 Demand, Supply, Equilibrium

Zhiyuan Chen

iMBA, Fall Semester 2025

RENMIN BUSINESS SCHOOL

Overview

- Consumer Demand
  - Demand Schedule/Curve
  - Demand Shifters
  - *Consumer Surplus
- Producer Supply
  - Supply Schedule/Curve
  - Supply Shifters
- Market Equilibrium

Demand

Perfectly Competitive Market

- A **market** is a group of buyers and sellers of a particular product
- A **competitive market** is one with many buyers and sellers, each having a negligible effect on price
  - wheat, mineral water
- In a **perfectly competitive** market（完全竞争市场）：
  - All goods are exactly the same
  - Buyers & sellers so numerous that no one can affect market price – each is a "**price taker（价格接受者）**"
- Currently, we assume markets are *perfectly competitive*

Perfectly Competitive Market

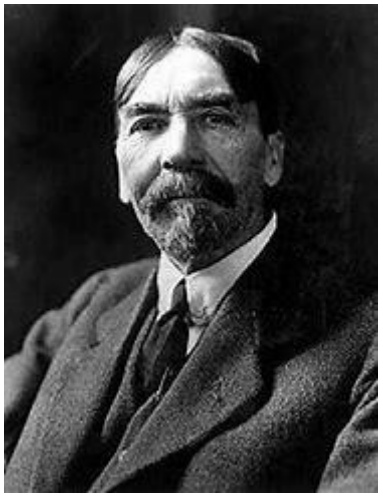
- The **quantity demanded** of any good is the amount of the good that buyers are willing and able to purchase.
- **Law of demand**（需求法则）: *Other things being equal* (ceteris paribus), the quantity demanded of a good falls when the price of the good rises.

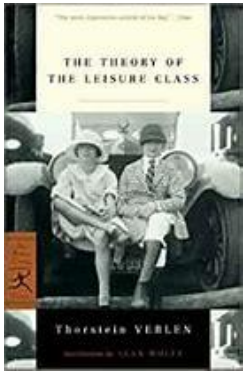


Can you find some real-life example that violates the law of demand?

Veblen Goods

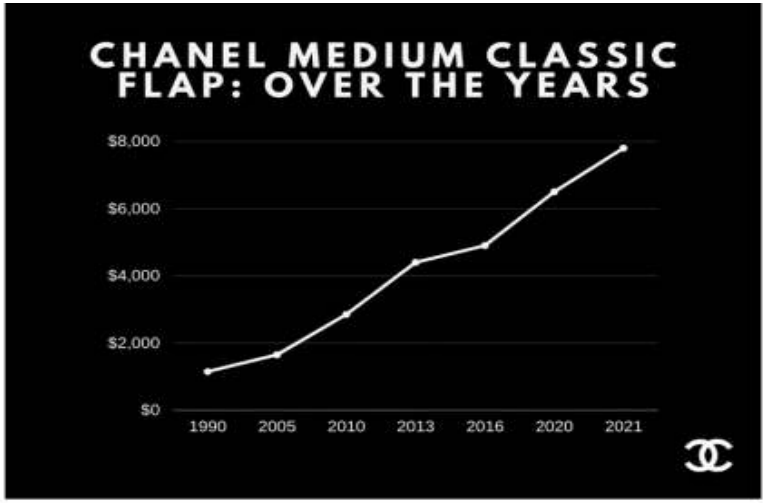
- Luxury items where *higher prices can lead to higher demand* because the goods serve as a status symbol.
  - luxury cars, designer handbags, and high-end wines
- For such goods, a higher price might indicate *exclusivity and desirability*, driving up demand.







Thorstein Veblen (1857-1929) Economist and Sociologist







The Demand Schedule

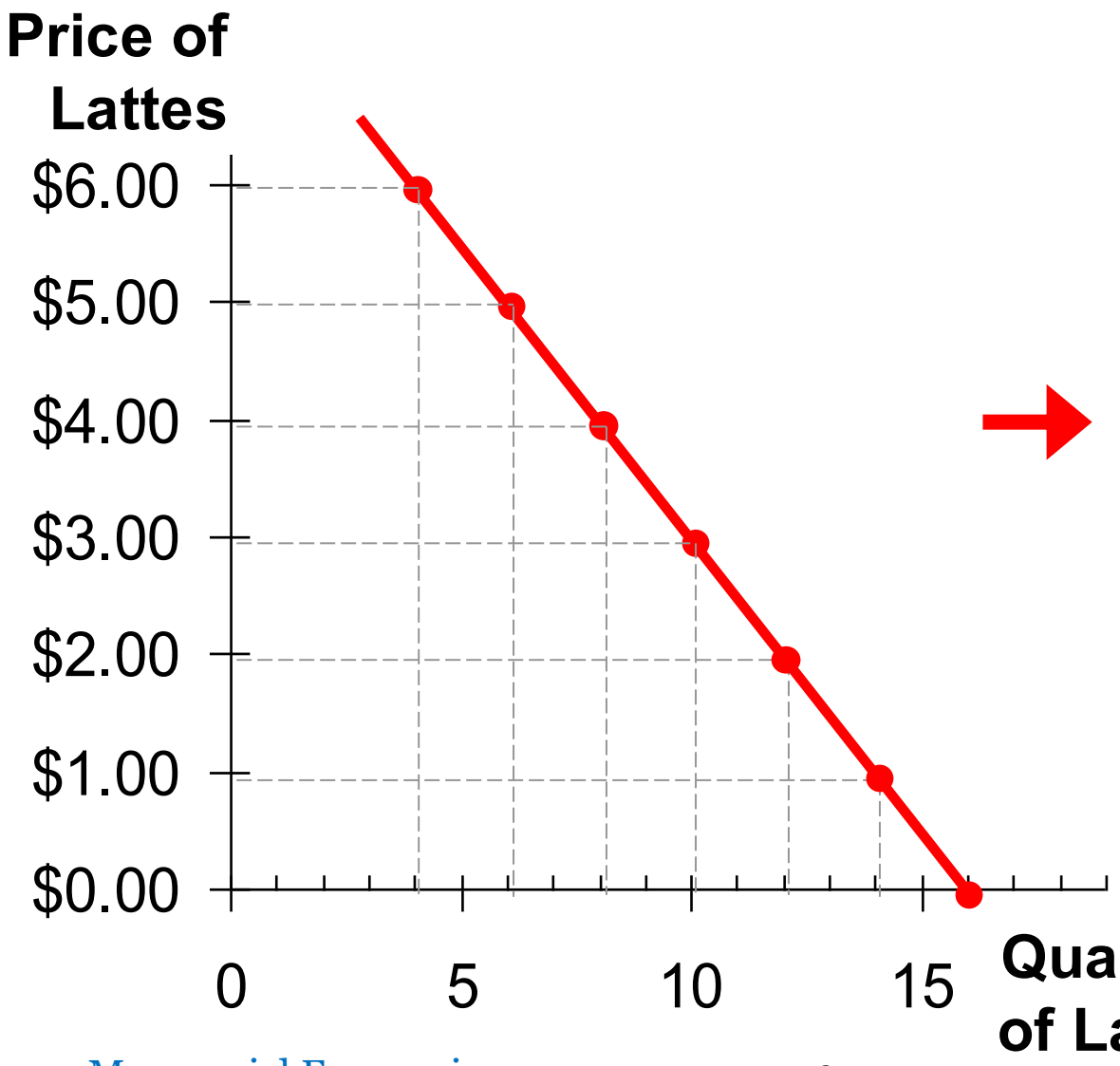
- **Demand Schedule (需求表):** a table that shows the relationship between the price of a good and the quantity demanded

- **Example:** Bob's demand for lattes in a week

- **Question:** Does the demand schedule follow the *law of demand*?

Price of lattes	Quantity of lattes demanded
$0.00	16
1.00	14
2.00	12
3.00	10
4.00	8
5.00	6
6.00	4

Bob's Demand Schedule & Curve



Price of lattes	Quantity of lattes demanded
$0.00	16
1.00	14
2.00	12
3.00	10
4.00	8
5.00	6
6.00	4

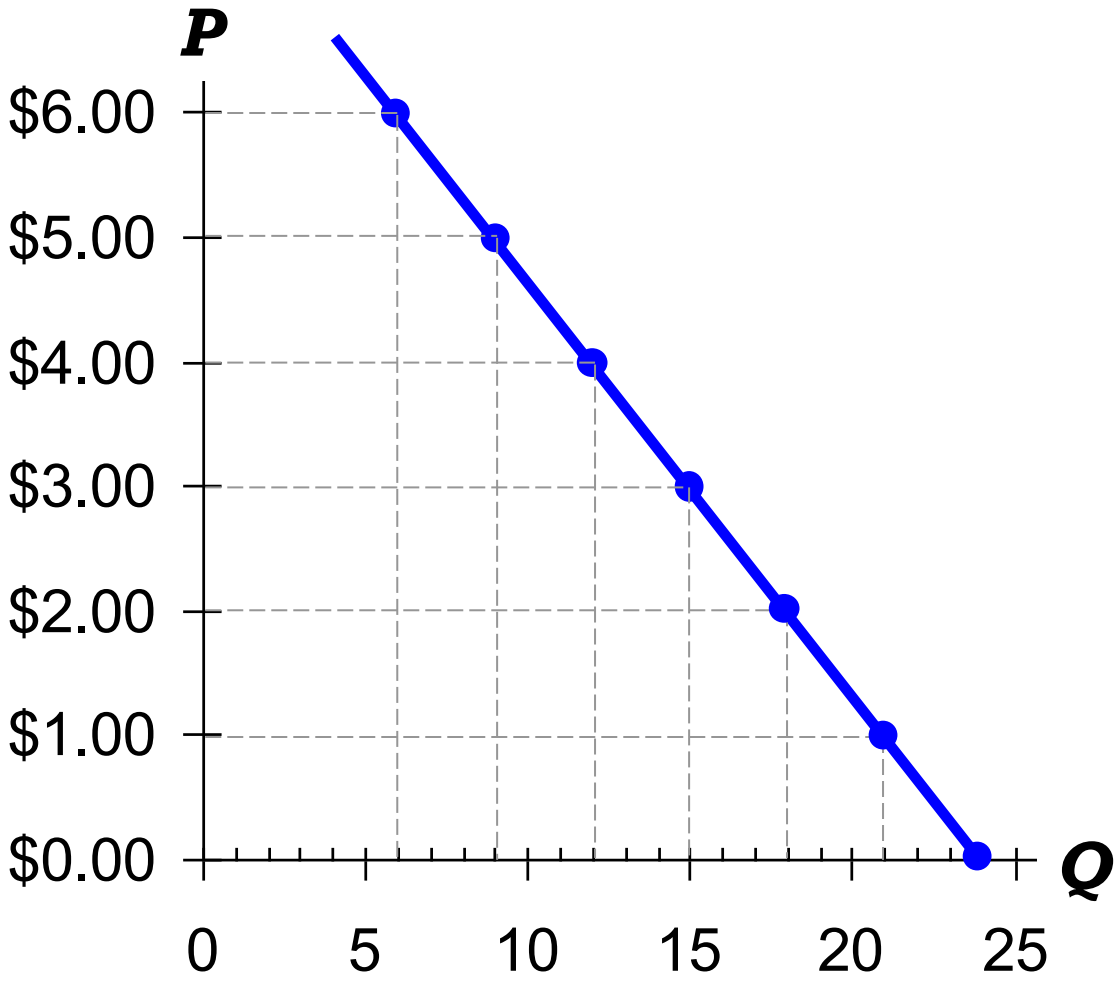
RENMIN BUSINESS SCHOOL

Individual Demand to Market Demand

- The quantity demanded in the market is the sum of the quantities demanded by all buyers at each price.
- Suppose Bob and Hellen are the only two buyers in the Latte market. (Q^d = quantity demanded)

Price	Bob's Q^d		Hellen's Q^d		Market Q^d
$0.00	16	+	8	=	24
1.00	14	+	7	=	21
2.00	12	+	6	=	18
3.00	10	+	5	=	15
4.00	8	+	4	=	12
5.00	6	+	3	=	9
6.00	4	+	2	=	6

Market Demand Schedule & Curve



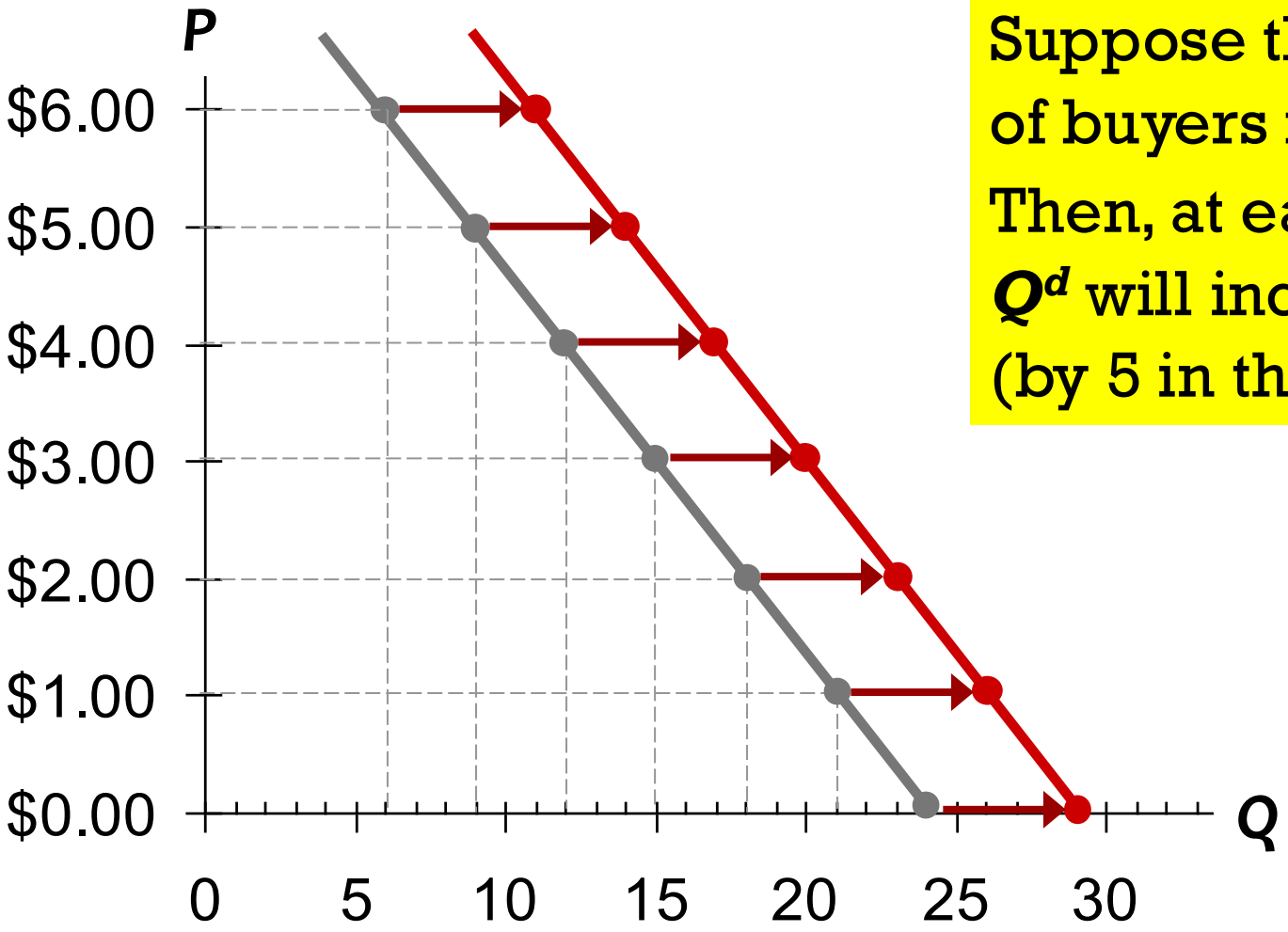
P	Q^d (Market)
$0.00	24
1.00	21
2.00	18
3.00	15
4.00	12
5.00	9
6.00	6

Demand Curve Shifters

- The demand curve shows how price affects quantity demanded, other things being equal.
- These "other things" are non-price determinants of demand (i.e., things that determine buyers' demand for a good, other than the good's price).
- *Changes in them shift the D curve…*
  - **Income**
    - Normal good
    - Inferior good
  - **Prices of Related Goods**
    - Prices of substitutes
    - Prices of complements
  - **Advertising and consumer tastes**
  - **Consumer expectations**
  - **Population**



Demand Curve Shifts: Increase # of Buyers



Suppose the number of buyers increases.

Then, at each ***P***, Q^d will increase (by 5 in this example)

Demand Curve Shifters: Increase Income

- Demand for a normal good （正常物品） is positively related to income.
  - Increase in income causes increase in quantity demanded at each price, shifts ***D*** curve to the right.
- Demand for an inferior good（低档物品） is negatively related to income, e.g., *instant noodles, pickles, low-quality rice, animal fat oil*
  - An increase in income shifts ***D*** curve for inferior goods to the left





D-Curve Shifters: Prices of Related Goods

- Two goods are substitutes（替代品） if an increase in the price of one causes an increase in demand for the other.
  - **Example:** Huawei Mate 60 Pro and iphone 15
- Two goods are complements（互补品） if an increase in the price of one causes a fall in demand for the other.
  - **Example:** Smartphones and Apps



Can you draw the demand curve for flight tickets and show how the expansion of high-speed railway affects the demand curve?

HSR and Demand for Flights



Eggs & Related Food Prices

- Related good price can differ in different occasions
  - •Eggs & bacon/sausage (in US and England)
  - •Eggs & tomato (in Northern China)





Demand Curve Shifters: Tastes

- Anything that causes a shift in tastes toward a good will increase demand for that good and shift its ***D*-curve** to the right
- **Example 1:** The Covid-19 pandemic has caused an increase demand for masks, disinfectants, video meeting, etc.
- **Example 2:** Preference for milk tea and coffee drinks has soared in the past decade

茶顏悅色 Modern China Tea Shop

HEEKCAA 喜茶



蜜雪冰城 始于1997·冰淇淋与茶

Demand Curve Shifters: Expectations

Expectations affect people's decisions.

- **Example 1:** If people expect their incomes to rise, their demand for meals at expensive restaurants may increase now.

- **Example 2:** In economic downturns, people are unsecure about their future job, demand for MBA education may increase.

Summary of Demand Determinants

Variable	A change in this variable…
Price	…causes a movement along the ***D*** curve
# of buyers	…shifts the ***D*** curve
Income	…shifts the ***D*** curve
Price of related goods	…shifts the ***D*** curve
Tastes	…shifts the ***D*** curve
Expectations	…shifts the ***D*** curve

Quiz: Demand Curve

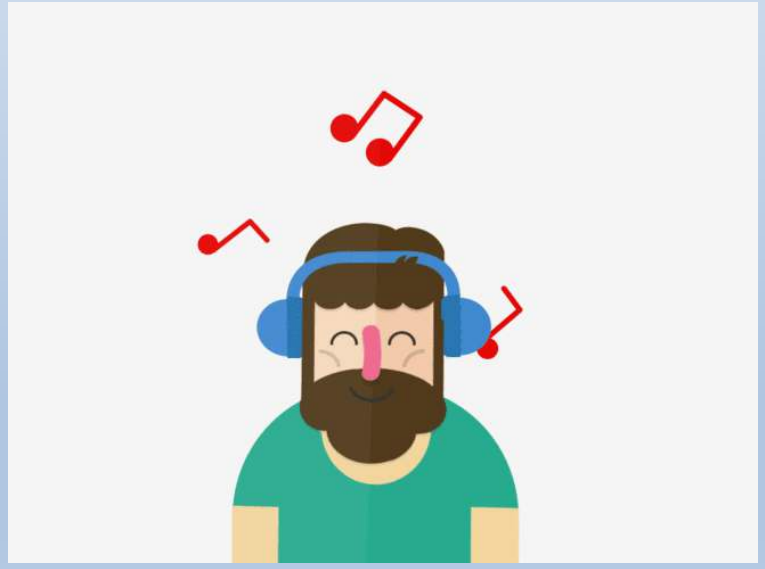
Draw a demand curve for *QQ/iTunes* Music memberships. What happens to it in each of the following scenarios?

Everyone draws diagrams!

A. The price of smart phones falls

B. The membership fee of QQ Music falls

C. The membership fee of NetEase Cloud (网易云) Music falls



Case Study: Demand Surge in Home Appliances

- Small home appliances see big growth in sales
  - Source: *China Daily*, 5 Oct 2020

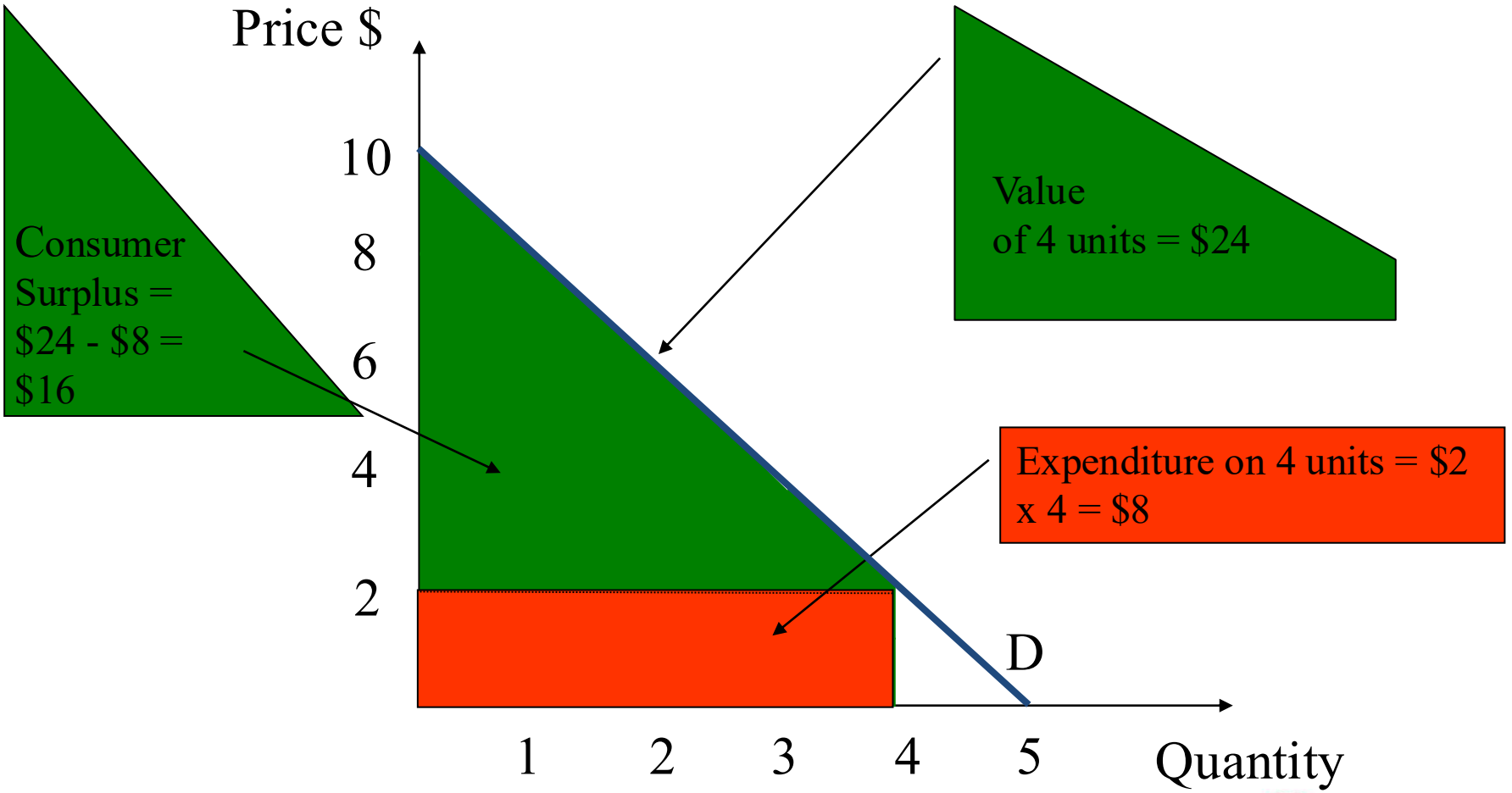
- **Questions**
  - *What have changed the demand of small home appliances in 2020?*
  - *Which appliance's sales revenue might have lost its momentum in 2021?*



Consumer Surplus

- The value consumers get from a good but do not have to pay for.
- Consumer surplus is particularly useful in marketing and other disciplines emphasizing strategies like *value pricing* and *price discrimination*.
- Consumer surplus is large → I got a great deal!
  - That company offers a lot of bang for the buck!
  - Dell provides good value.
- Consumer surplus is low → I got a lousy deal!
  - That car dealer drives a hard bargain!
  - I almost decided not to buy it!
  - They tried to squeeze the very last cent from me!

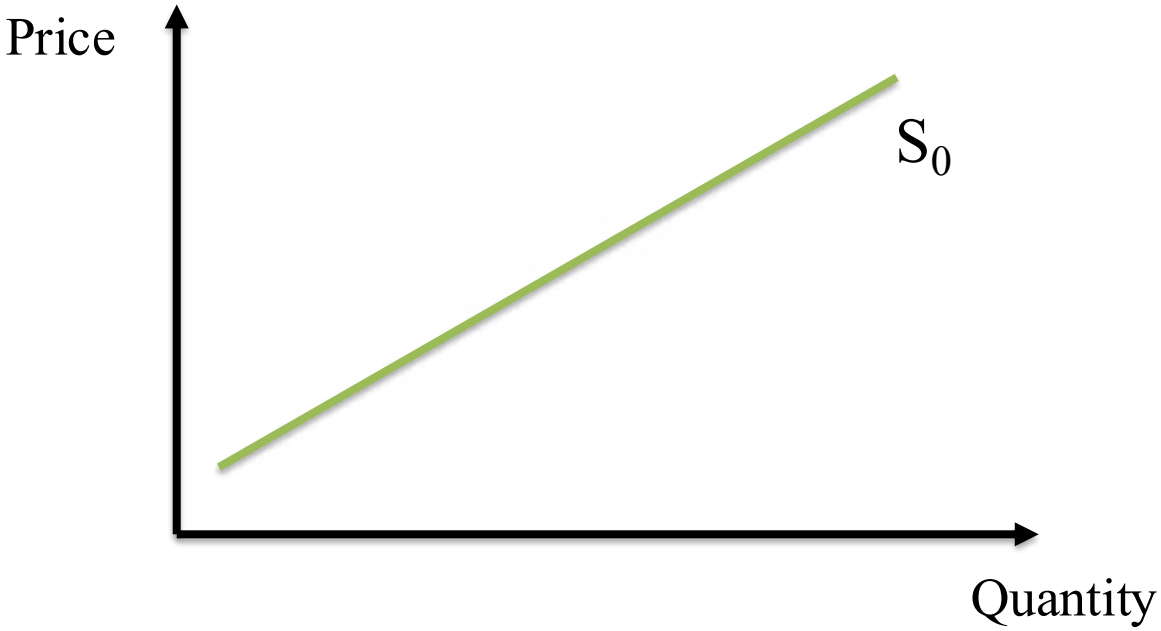
Consumer Surplus: Graphic Illustration



Supply

Supply Curve

- The supply curve shows the amount of a good that will be produced at alternative prices.
- *Law of Supply*
  - The supply curve is upward sloping.



Supply Shifters



- Input prices
- Technology or government regulations
- Number of firms
  - Entry
  - Exit
- Substitutes in production
- Taxes
- Producer expectations

RENMIN BUSINESS SCHOOL

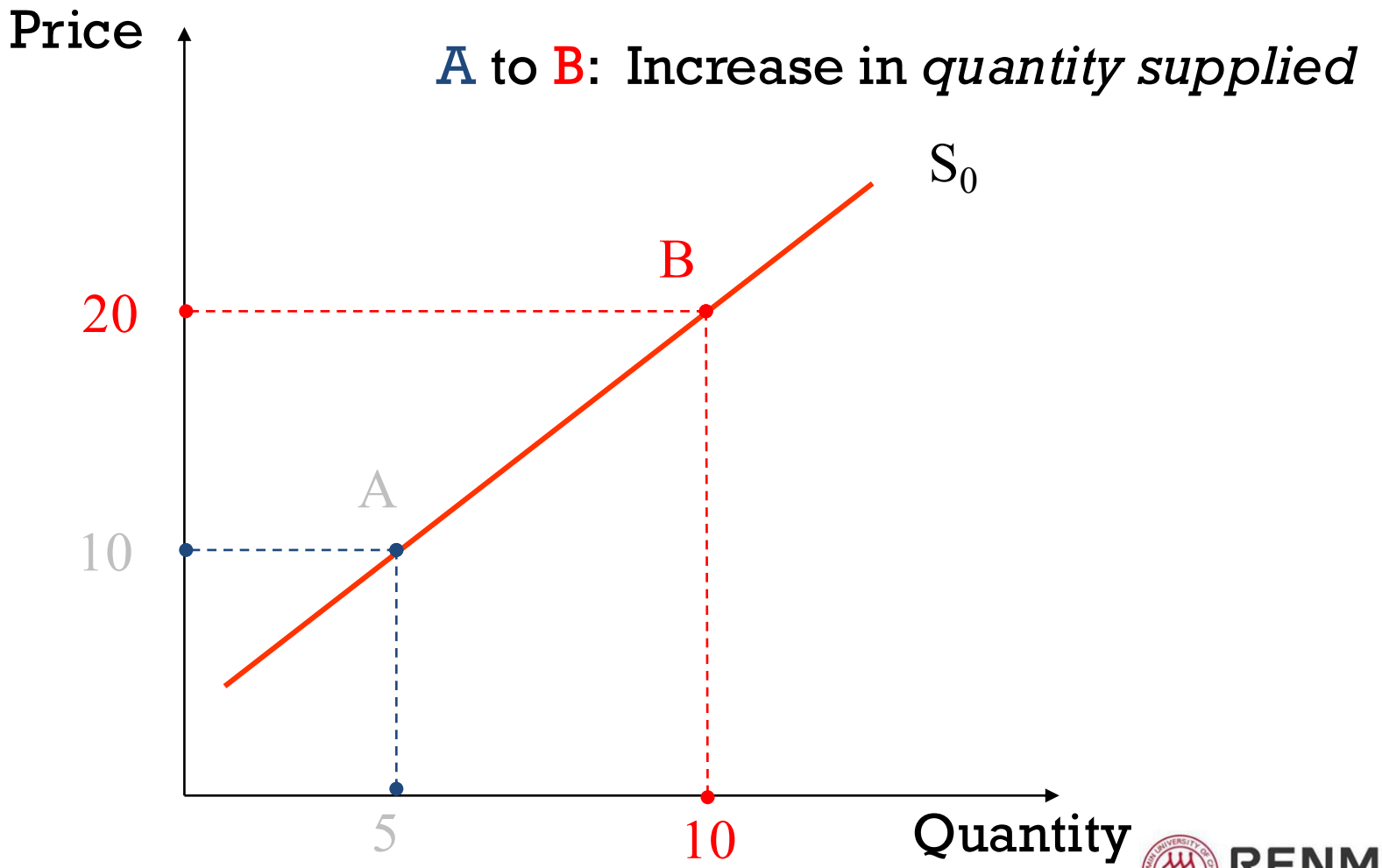
The Supply Function

- An equation representing the supply curve:

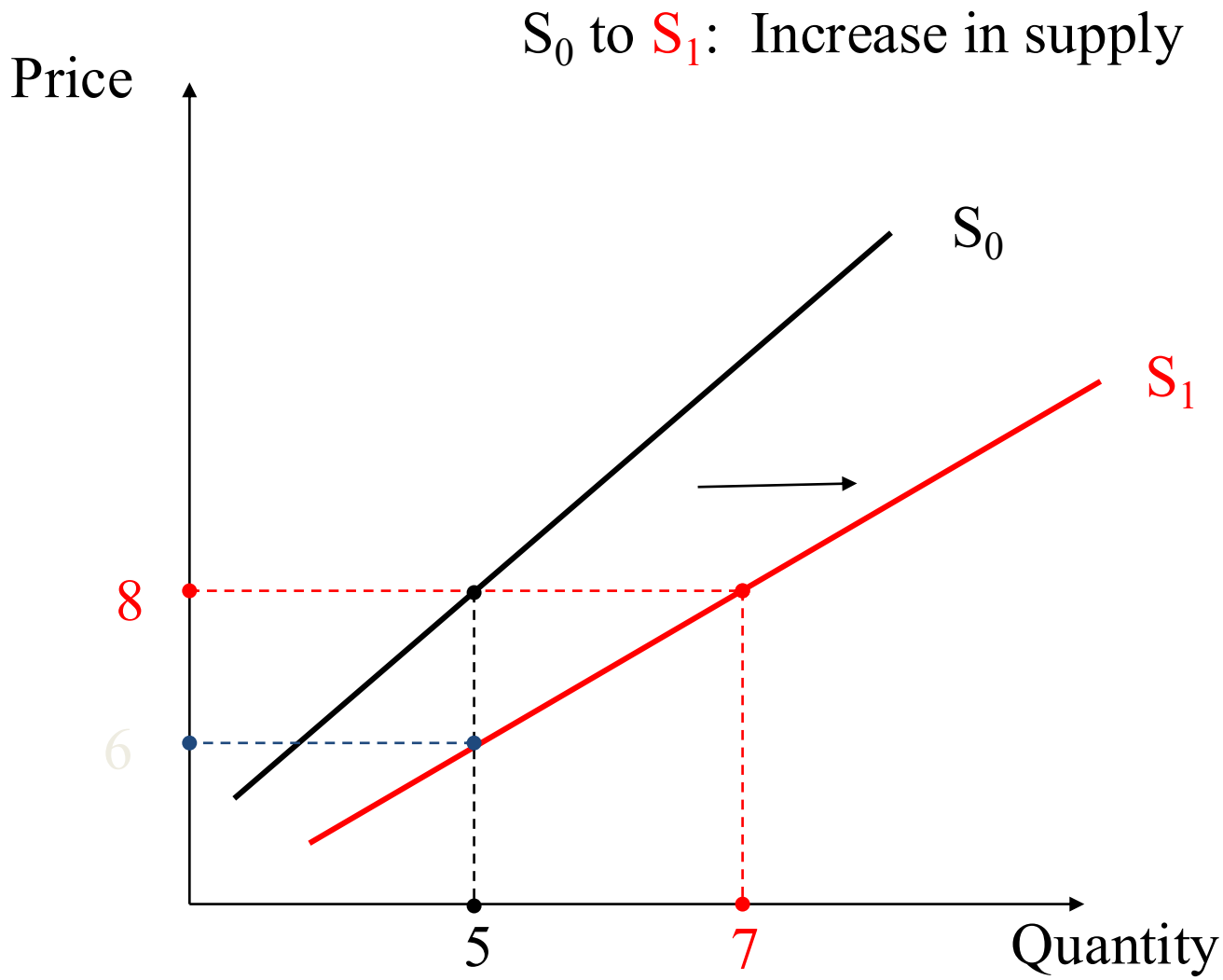
$$Q_x^S = f(P_x, W, H)$$

- Q_x^S = quantity supplied of good X.
- P_x = price of good X.
- W = price of inputs (e.g., wages).
- H = other variable affecting supply.

Changes in Quantity Supplied

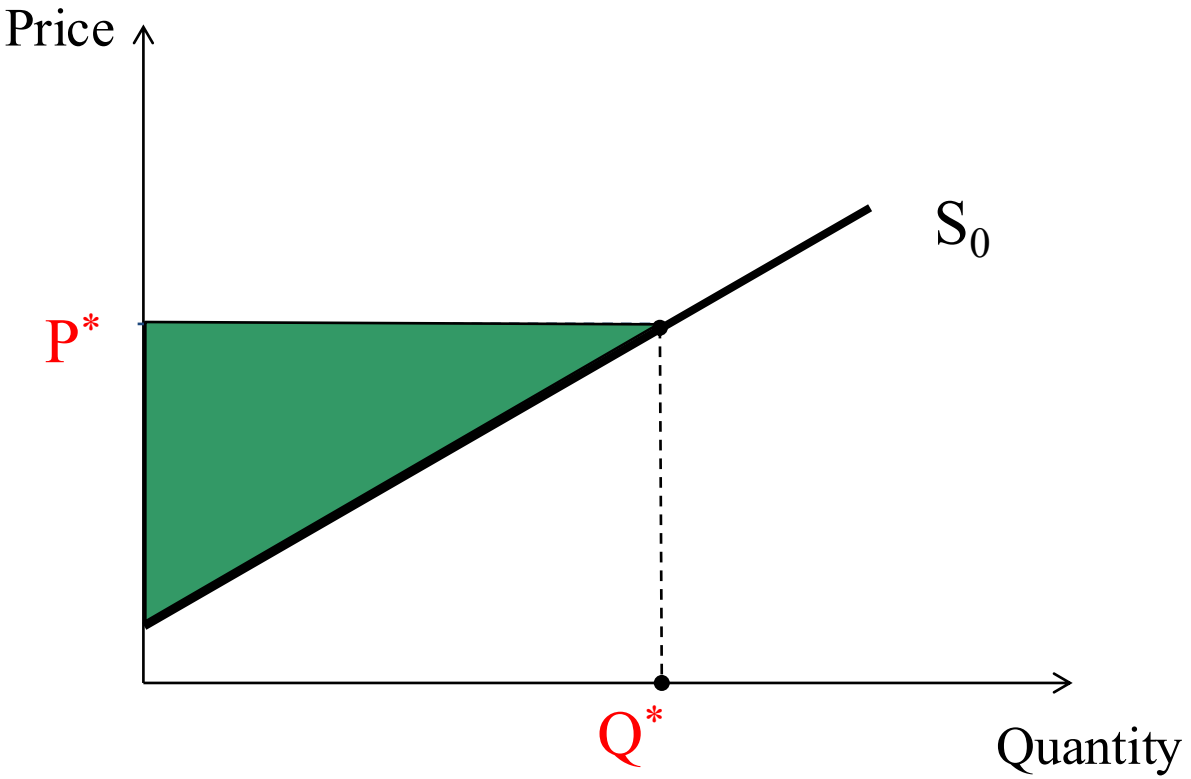


Supply Curve Shifts



Producer Surplus

- The amount producers receive in excess of the amount necessary to induce them to produce the good.



Practice: Supply Curve Shifts

Draw a supply curve for *Starbucks* coffee. What happens to it in each of the following scenarios?

Everyone draws diagrams

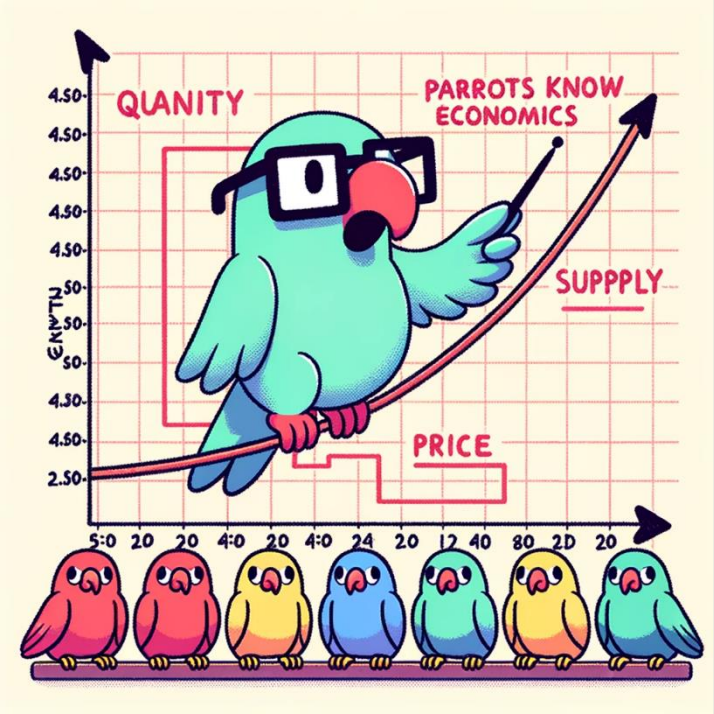
A. The price of *Luckin* coffee decreases

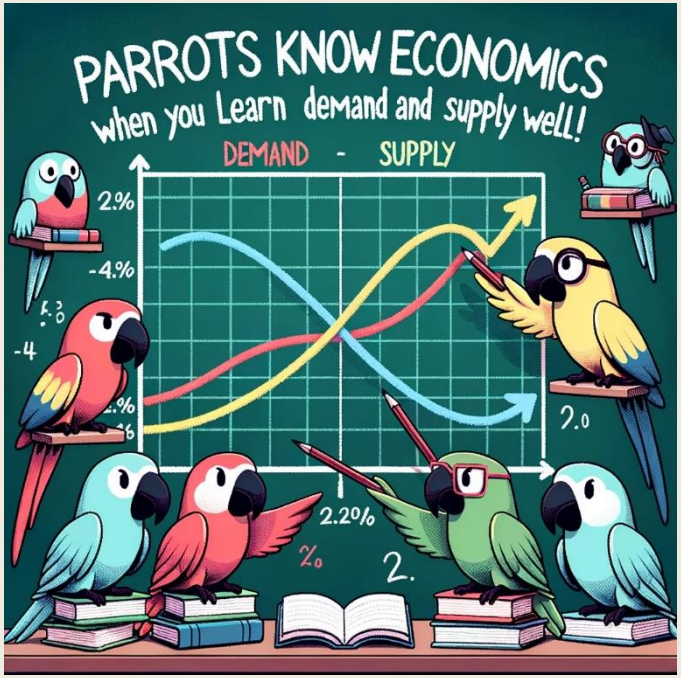
B. The price of Starbucks Coffee increases

C. Rent of commercial property increases



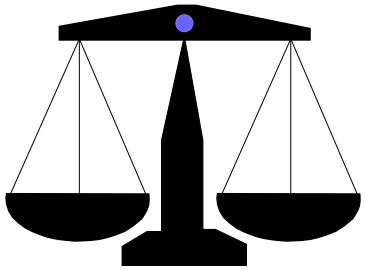
Market Equilibrium





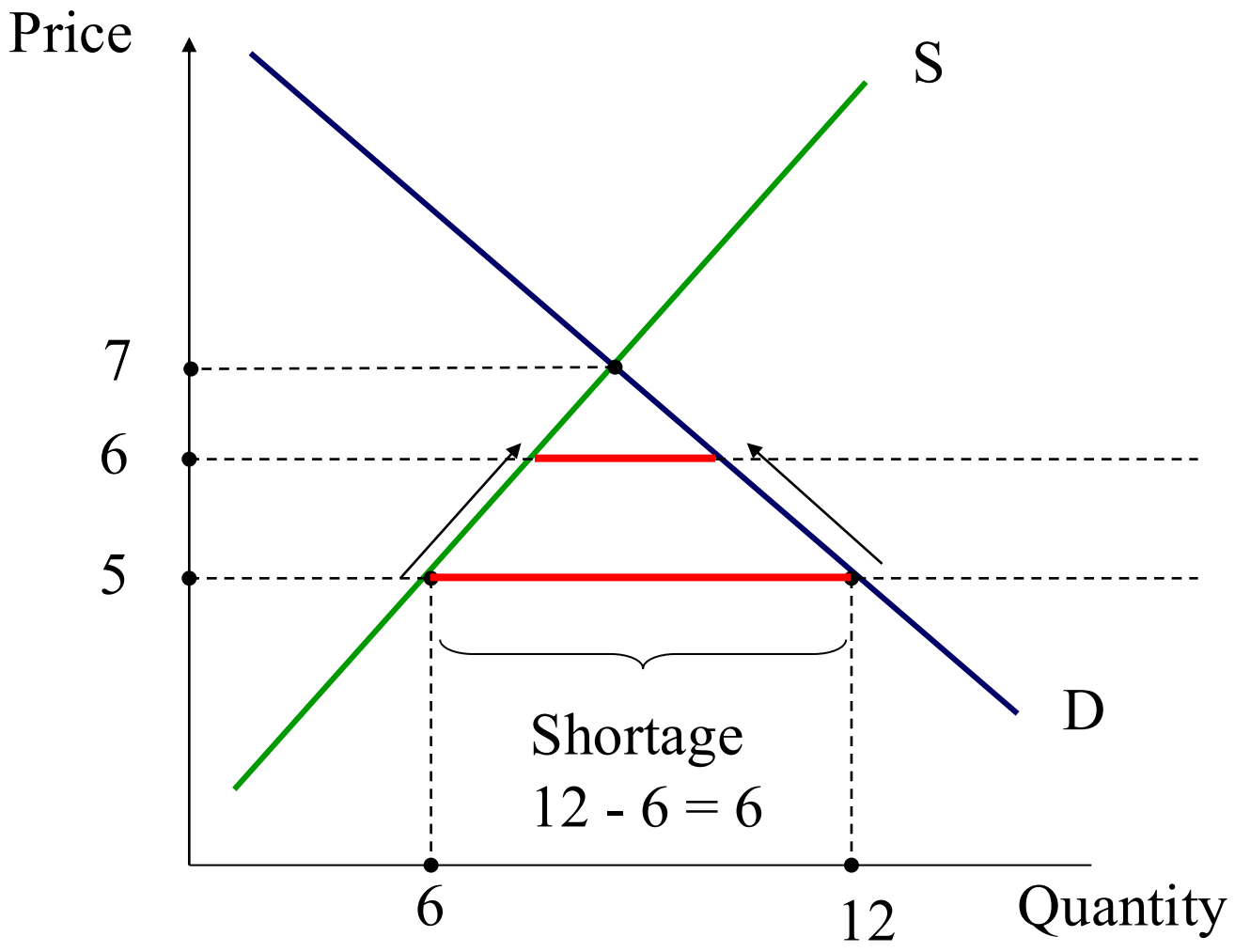
Market Equilibrium

- The Price (P) that balances supply and demand
  - $Q_x^S = Q_x^d$
  - No shortage or surplus

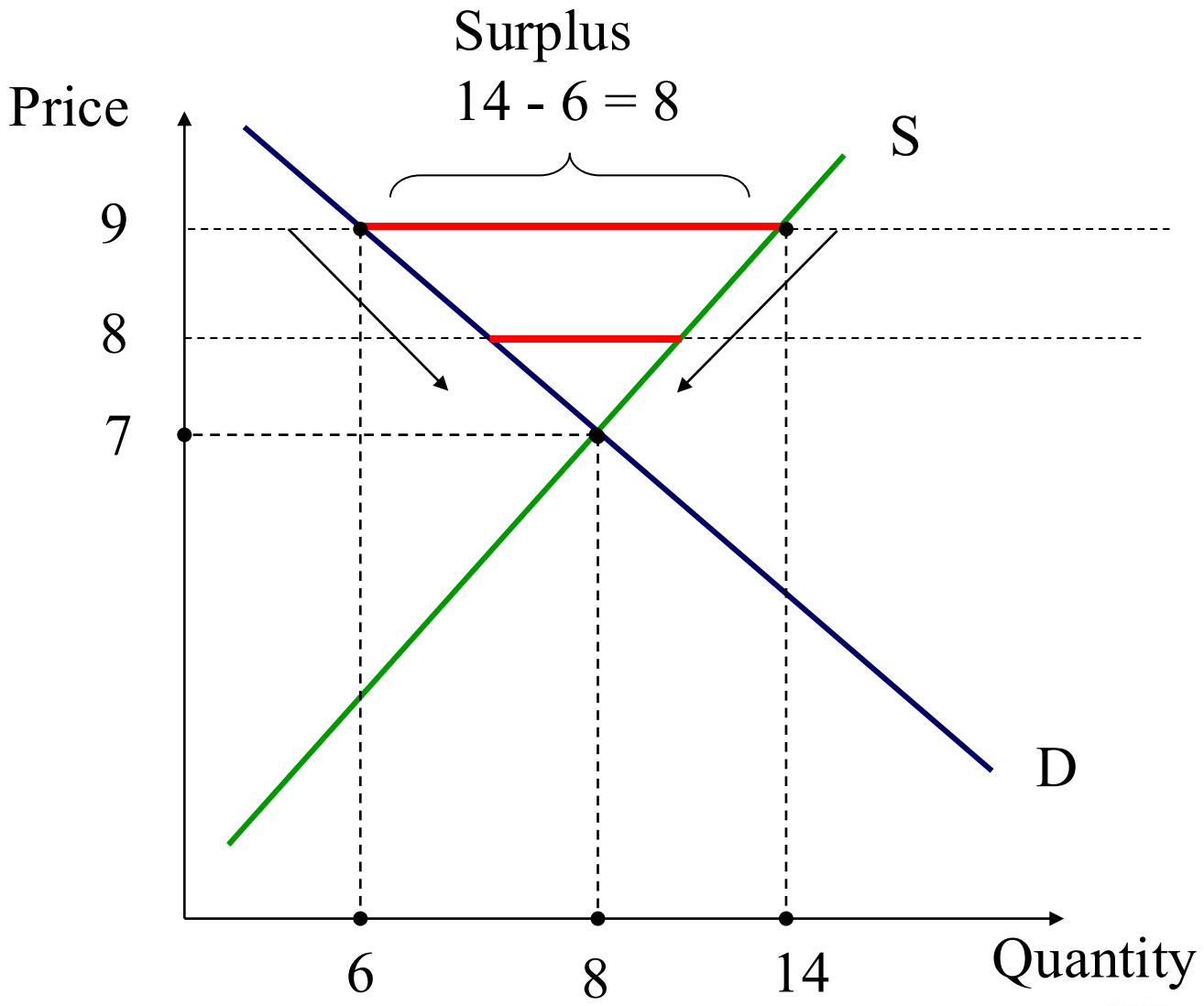


- Market forces drive the price and quantity to the market equilibrium:
  - **Shortage** ($Q_x^S < Q_x^d$)**:** excessive demand drives up price
  - **Surplus** ($Q_x^S > Q_x^d$)**:** excessive supply drives down price

If price is too low…Shortage



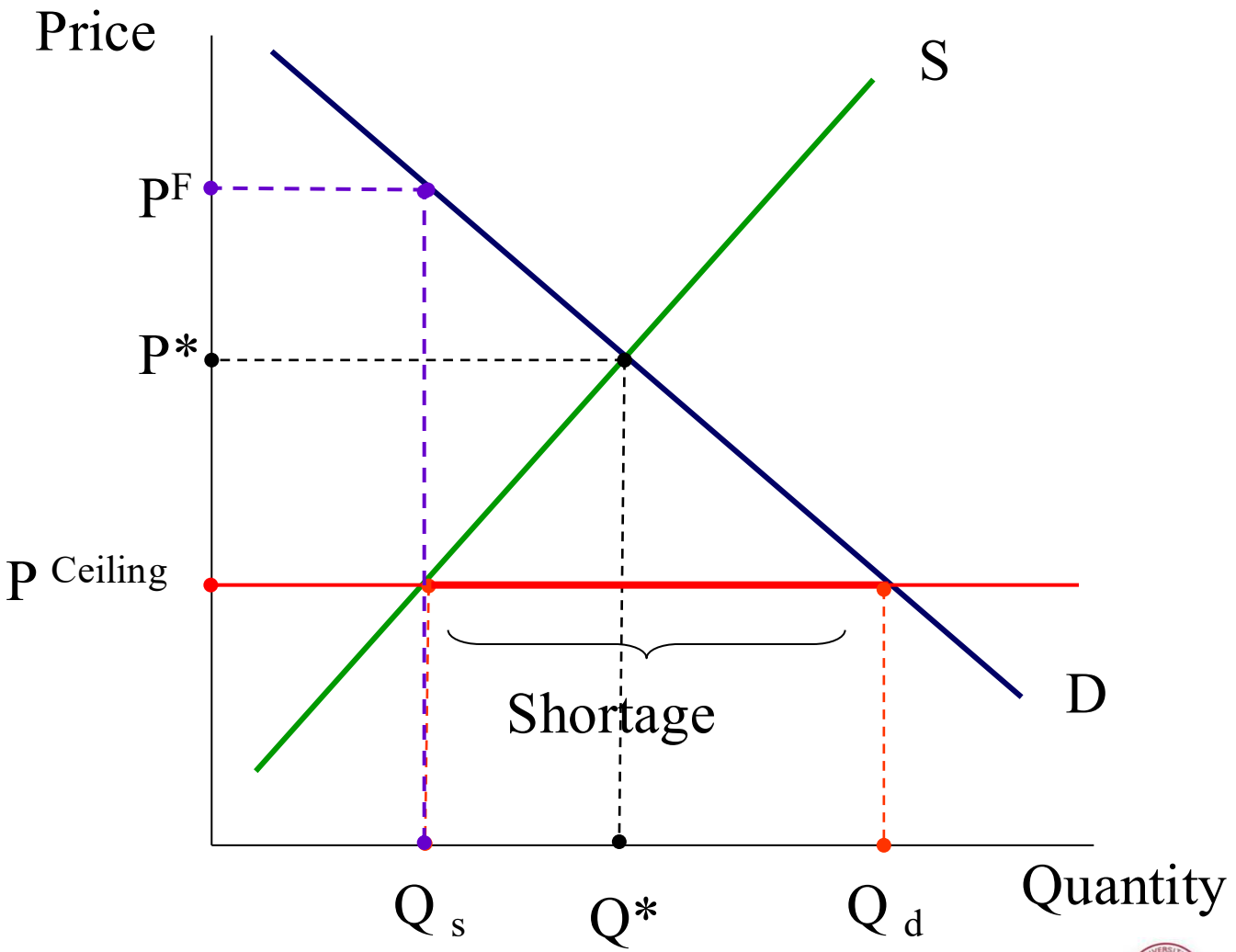
If price is too high…Surplus



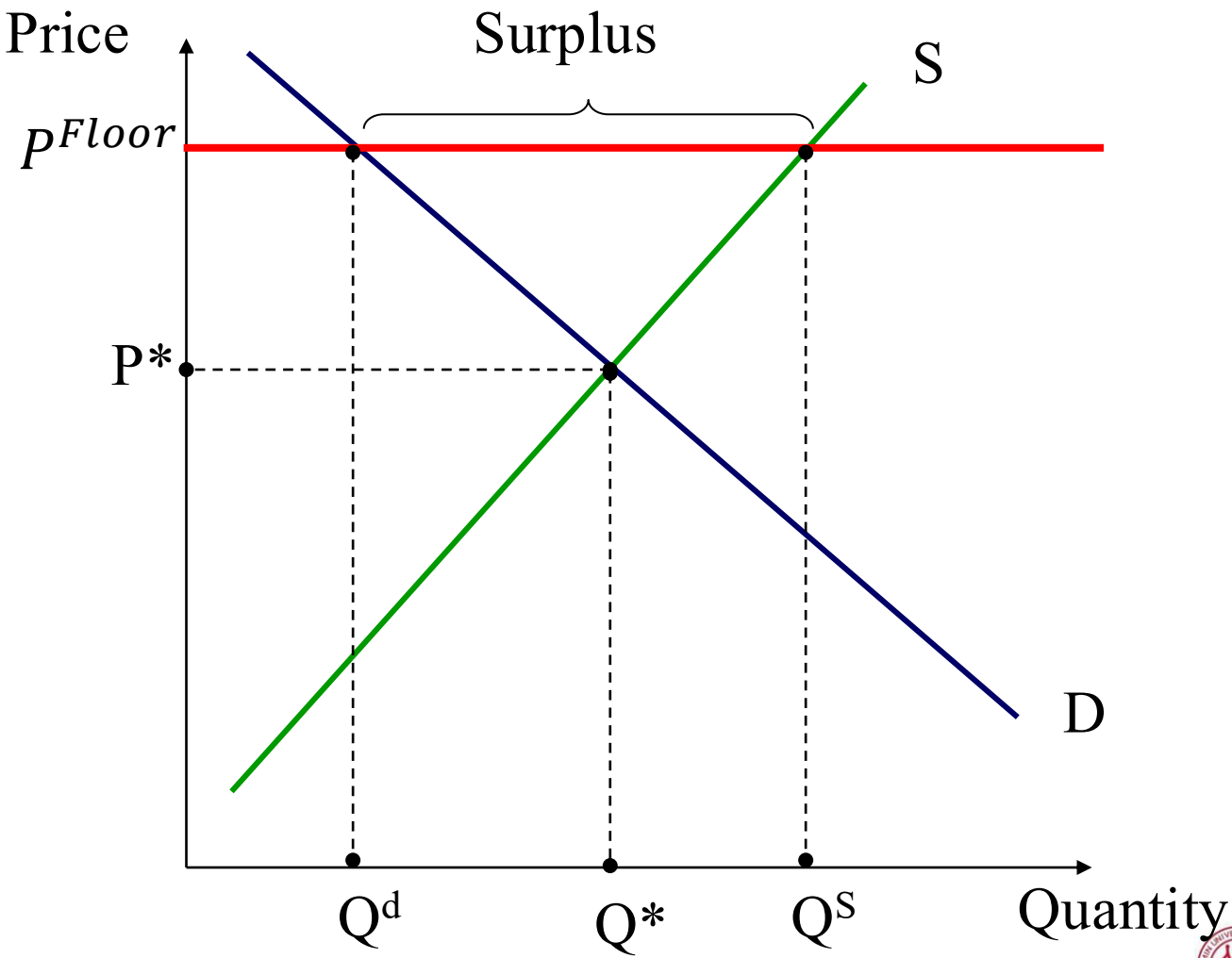
Price Restrictions

- Price Ceilings
  - The *maximum* legal price that can be charged.
  - Examples:
    (1) Gasoline prices in the 1970s.
    (2) Housing Rents in Beijing.
    (3) Proposed restrictions on ATM fees.
- Price Floors
  - The *minimum* legal price that can be charged.
  - Examples:
    (1) Minimum wage.
    (2) Agricultural price supports.

Impact of a Price Ceiling

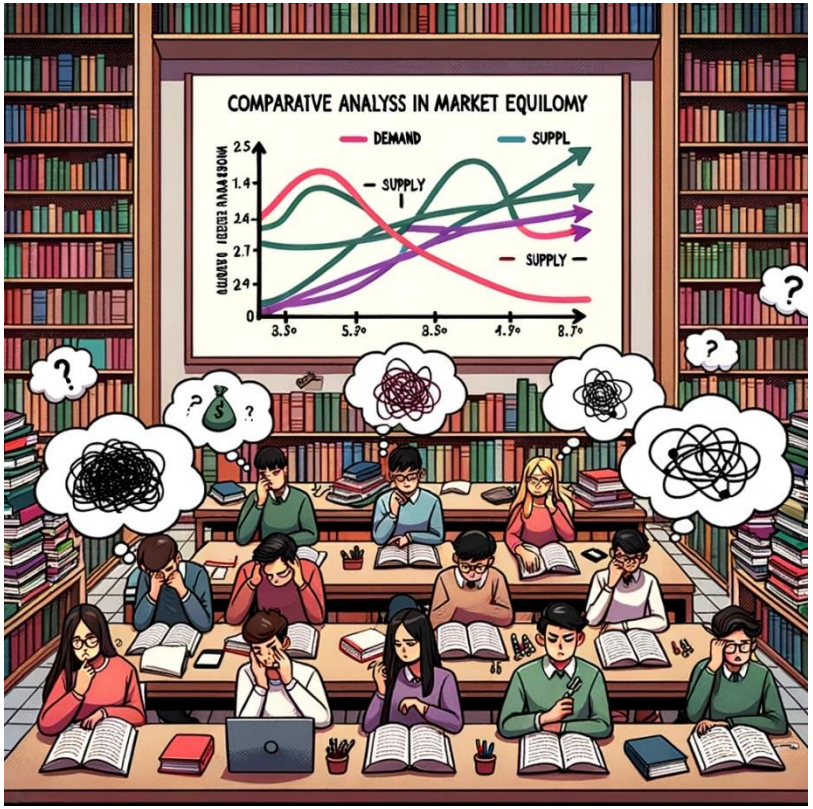


Impact of a Price Floor



Comparative Static Analysis

- How do the equilibrium price and quantity change when a determinant of supply and/or demand change?



Case Study: Market Equilibrium Analysis

- **Event:** *China Daily* reports that the prices of PC memory chips are expected to fall by 5-8 percent over the next six months.

1. *Scenario 1:* You manage a small firm that manufactures PCs
2. *Scenario 2:* You manage a small software company



Everybody draws a diagram to show the change in market equilibrium in two scenarios, and think about action plans for the firm.

Revisit the Case Study on Home Appliances

- Small home appliances see big growth in sales
  - Source: *China Daily*, 5 Oct 2020

- What is the long-run impact of this demand change?

Social Welfare and ESG

Social Welfare = Total Surplus

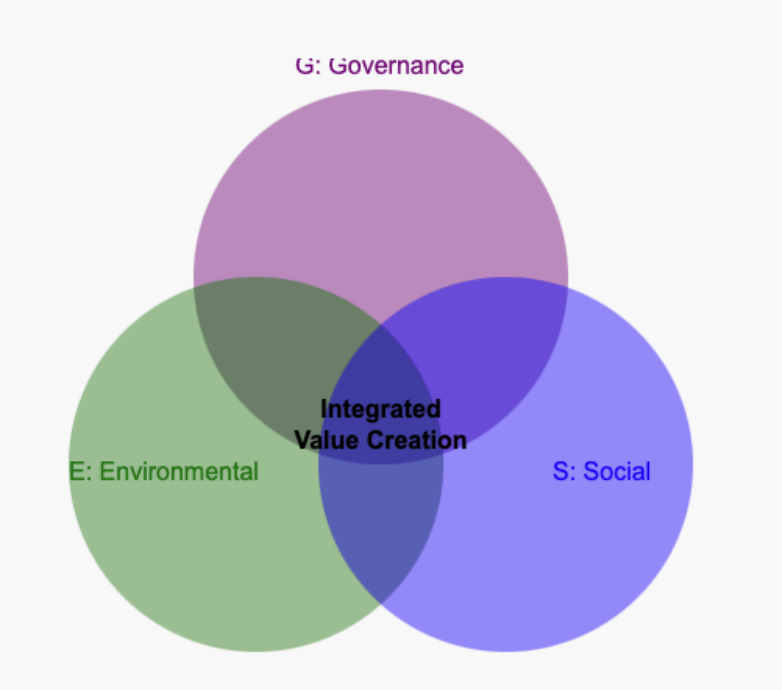
- **Total Surplus = Consumer Surplus + Supplier Surplus**
- **Social Welfare** increases when:
  - Consumers gain more surplus
  - Suppliers gain more surplus
- **Efficient Equilibrium** maximizes the total surplus
- Maket equilibirium is *not necessarily* the effcient equilibrium:
  - **Externalities**: pollution, innovation
  - **Public goods**: national defense, open-source software
  - **Market power**: patents
  - **Information failures**: used-goods market (闲鱼)

Foundations of ESG

- **Environmental (E)**: Efficient use of natural resources, reducing emissions, sustainable practices.

- **Social (S)**: Fair labor, diversity & inclusion, community engagement, product safety.

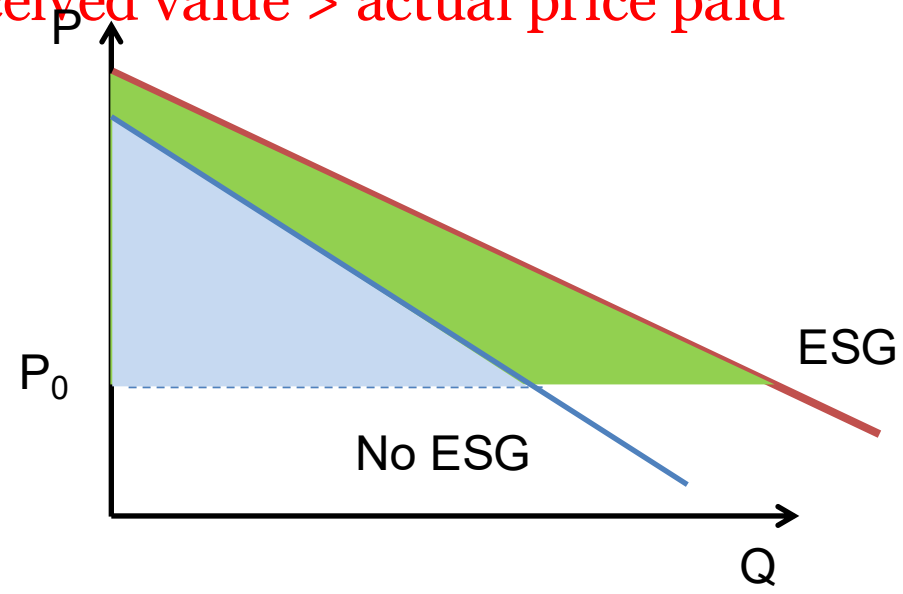
- **Governance (G)**: Transparency, ethical leadership, compliance, accountability to stakeholders.



ESG on Consumer Surplus

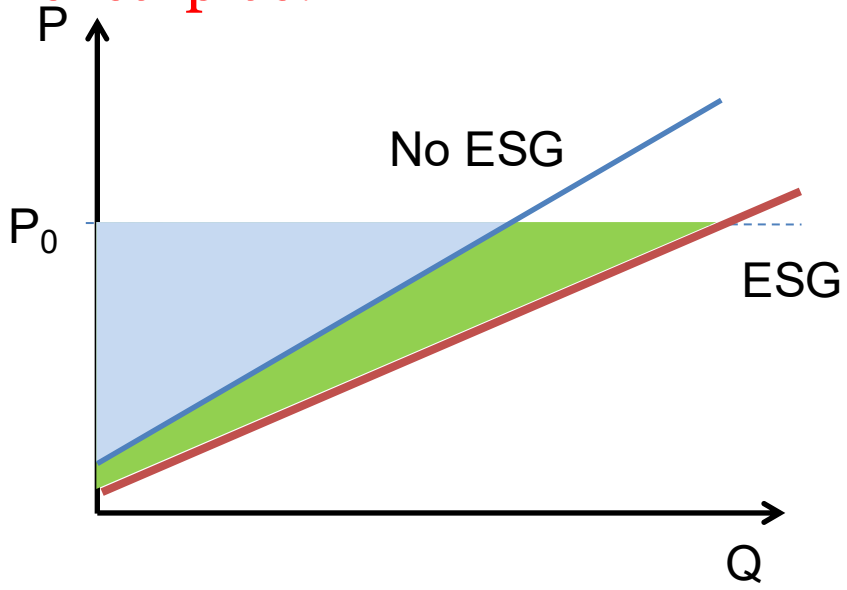
- ESG initiatives increase consumers' **willingness to pay** by enhancing *trust, quality perception, and brand reputation*.
- Environmentally friendly and socially responsible products create **additional value** for consumers beyond functional benefit.
- Expanded consumer surplus because the:

perceived value > actual price paid



ESG on Supplier Surplus

- ESG practices improve **operational efficiency** (e.g., sustainable resource use, energy savings).
- They enhance access to capital and investor trust, **lowering financing costs**.
- Suppliers differentiate through ESG, allowing **premium pricing and stronger customer loyalty**.
- Greater supplier surplus?



Example of an ESG Initiative

A consumer goods company introduces biodegradable packaging.

- **Environmental (E)**: Reduces plastic waste and carbon footprint.
- **Social (S)**: Improves community health and public image.
- **Governance (G)**: Transparent reporting on sustainability goals and compliance with regulations.

Impact Linkages:

- **Consumer Surplus**: Customers value environmentally safe products more → higher willingness to pay.
- **Supplier Surplus:** Operational efficiencies (reduced waste disposal costs) + brand premium → higher profitability.

Summary

- Demand Analysis:
  - Law of demand
  - Demand schedule/Demand Curve
  - Demand Shifters
- Supply
  - Law of supply
  - Supply curve
  - Supply shifters
- Market equilibrium: comparative statics
- Social welfare